

Index.aspx

```
<%@ Page Language="C#" AutoEventWireup="true" %>

<!DOCTYPE html>
<html lang="en">
<head runat="server">
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Lab 7 - Customer Segmentation</title>
    <style>
        .linkStyle {
            display: block;
            margin: 10px 0;
            font-size: 16px;
        }
        body {
            font-family: Arial, sans-serif;
            margin: 40px;
            background-color: #f5f5f5;
        }
        .container {
            max-width: 700px;
            margin: auto;
            background: white;
            padding: 30px;
            border: 1px solid #ddd;
            border-radius: 8px;
        }
        h2 {
            color: #333;
            border-bottom: 2px solid #4CAF50;
            padding-bottom: 10px;
        }
        .footer {
            margin-top: 30px;
            padding-top: 15px;
            border-top: 1px solid #ddd;
            text-align: center;
            font-size: 12px;
            color: #666;
        }
    </style>
</head>
<body>
    <form id="form1" runat="server">
        <div class="container">
            <h2>Lab 07 Customer Segmentation </h2>

            <asp:HyperLink ID="lnkSegmentation" runat="server"
                NavigateUrl="Default.aspx"
                CssClass="linkStyle">
                Demonstration Page
            </asp:HyperLink>

            <asp:HyperLink ID="lnkProblem" runat="server"
                NavigateUrl="Problem_Statement.txt"
                CssClass="linkStyle">
                Problem Statement
            </asp:HyperLink>
        </div>
    </form>
</body>
</html>
```

```

        <asp:HyperLink ID="lnkReport" runat="server"
            NavigateUrl="Code_Implementation_Report.pdf"
            CssClass="linkStyle">
            Code Implementation Report
        </asp:HyperLink>

        <asp:HyperLink ID="lnkOutput" runat="server"
            NavigateUrl="Sample_Output.pdf"
            CssClass="linkStyle">
            Sample Output
        </asp:HyperLink>

    </div>
</form>
</body>
</html>

```

Index.aspx.cs

```

using System;
using System.Collections.Generic;
using System.Linq;
using System.Web;
using System.Web.UI;
using System.Web.UI.WebControls;

namespace IADLAB07
{
    public partial class index : System.Web.UI.Page
    {
        protected void Page_Load(object sender, EventArgs e)
        {

        }
    }
}

```

Default.aspx

```

<%@ Page Language="C#" AutoEventWireup="true" %>

<!DOCTYPE html>

<html xmlns="http://www.w3.org/1999/xhtml">
<head runat="server">
    <title>Lab 7 - Customer Segmentation</title>
    <style>
        body {
            font-family: Arial, sans-serif;
            margin: 50px;
            background-color: #f5f5f5;
        }
        .container {
            max-width: 500px;
            margin: auto;
            background: white;

```

```

        padding: 30px;
        border: 1px solid #ddd;
        border-radius: 10px;
        text-align: center;
    }
    h1 {
        color: #4CAF50;
        border-bottom: 2px solid #4CAF50;
        padding-bottom: 10px;
    }
    .link {
        display: block;
        padding: 12px 20px;
        margin: 15px 0;
        background: #4CAF50;
        color: white;
        text-decoration: none;
        border-radius: 5px;
        font-size: 16px;
    }
    .link:hover {
        background: #45a049;
    }
</style>
</head>
<body>
    <form id="form1" runat="server">
        <div class="container">
            <h1> Lab 7 - Customer Segmentation</h1>

            <a href="CustomerSegmentation.aspx" class="link">
                Customer Segmentation Analysis
            </a>

            <a href="TargetedMarketing.aspx" class="link">
                Targeted Marketing Automation
            </a>

            <hr />

        </div>
    </form>
</body>
</html>

```

Default.aspx.cs

```

using System;
using System.Web.UI;

namespace IADLAB07
{
    public partial class Default : System.Web.UI.Page
    {
        protected void Page_Load(object sender, EventArgs e)
        {
            // Nothing needed here
        }
    }
}

```

```
}
```

CustomerSegmentation.aspx

```
<%@ Page Language="C#" AutoEventWireup="true" CodeBehind="CustomerSegmentation.aspx.cs"
Inherits="CustomerSegmentation" %>
```

```
<!DOCTYPE html>
<html xmlns="http://www.w3.org/1999/xhtml">
<head runat="server">
    <title>Customer Segmentation Analysis</title>
    <style>
        body {
            font-family: Arial, sans-serif;
            margin: 20px;
            background-color: #f5f5f5;
        }

        .container {
            max-width: 1300px;
            margin: auto;
            background: white;
            padding: 20px;
            border: 1px solid #ddd;
            border-radius: 8px;
        }

        h2 {
            color: #333;
            margin-top: 0;
            padding-bottom: 10px;
            border-bottom: 2px solid #4CAF50;
            text-align: center;
        }

        .stats {
            background: #e8f5e9;
            padding: 15px;
            margin: 20px 0;
            border: 1px solid #c8e6c9;
            border-radius: 5px;
            display: flex;
            justify-content: space-around;
            flex-wrap: wrap;
        }

        .stat-card {
            text-align: center;
            padding: 10px 20px;
            background: white;
            border-radius: 5px;
            min-width: 150px;
        }

        .stat-number {
            font-size: 24px;
            font-weight: bold;
            color: #4CAF50;
        }
    </style>
</head>
<body>
```

```

.stat-label {
    font-size: 12px;
    color: #666;
    margin-top: 5px;
}

.segment {
    margin-bottom: 30px;
    border: 1px solid #ddd;
    background: white;
    border-radius: 5px;
    overflow: hidden;
}

.segment-title {
    background: #4CAF50;
    color: white;
    padding: 12px 15px;
    font-weight: bold;
    font-size: 16px;
}

.segment-desc {
    font-size: 12px;
    font-weight: normal;
    margin-left: 10px;
    opacity: 0.9;
}

table {
    width: 100%;
    border-collapse: collapse;
}

th {
    background: #f2f2f2;
    padding: 10px;
    text-align: left;
    border: 1px solid #ddd;
    font-weight: bold;
}

td {
    padding: 8px 10px;
    border: 1px solid #ddd;
}

tr:hover {
    background-color: #f9f9f9;
}

.btn {
    padding: 8px 20px;
    margin: 5px;
    cursor: pointer;
    border: none;
    background: #4CAF50;
    color: white;
    border-radius: 4px;
}

```

```

        .btn-back {
            background: #666;
        }

        .btn:hover {
            background: #45a049;
        }

        .footer {
            text-align: center;
            margin-top: 20px;
            padding-top: 20px;
            border-top: 1px solid #ddd;
            color: #666;
            font-size: 12px;
        }

        .refresh-btn {
            text-align: right;
            margin-bottom: 15px;
        }

        .button-group {
            text-align: center;
            margin-top: 20px;
        }
    </style>
</head>
<body>
    <form id="form1" runat="server">
        <div class="container">
            <h2> Customer Segmentation Analysis</h2>

            <div class="refresh-btn">
                <asp:Button ID="btnRefresh" runat="server" Text=" Refresh Data"
                    CssClass="btn" OnClick="btnRefresh_Click" />
            </div>

            <div class="stats">
                <div class="stat-card">
                    <div class="stat-number"><asp:Label ID="lblTotalCustomers"
runat="server" Text="0" /></div>
                    <div class="stat-label">Total Customers</div>
                </div>
                <div class="stat-card">
                    <div class="stat-number"><asp:Label ID="lblActiveCustomers"
runat="server" Text="0" /></div>
                    <div class="stat-label">Active Customers</div>
                </div>
                <div class="stat-card">
                    <div class="stat-number"><asp:Label ID="lblInactiveCustomers"
runat="server" Text="0" /></div>
                    <div class="stat-label">Inactive/New Customers</div>
                </div>
                <div class="stat-card">
                    <div class="stat-number">${<asp:Label ID="lblTotalRevenue"
runat="server" Text="0" /></div>
                    <div class="stat-label">Total Revenue</div>
                </div>
            </div>
        </div>
    </form>
</body>
</html>

```

```

        <!-- Premium Customers -->
        <div class="segment">
            <div class="segment-title"> Premium Customers <span class="segment-
desc">(Total spent more than $2,000)</span></div>
            <asp:GridView ID="gvPremium" runat="server" AutoGenerateColumns="True"
                EmptyDataText="No premium customers found" Width="100%" />
        </div>

        <!-- Frequent Customers -->
        <div class="segment">
            <div class="segment-title"> Frequent Customers <span class="segment-
desc">(More than 3 orders)</span></div>
            <asp:GridView ID="gvFrequent" runat="server" AutoGenerateColumns="True"
                EmptyDataText="No frequent customers found" Width="100%" />
        </div>

        <!-- Bulk Buyers -->
        <div class="segment">
            <div class="segment-title"> Bulk Buyers <span class="segment-
desc">(Average quantity more than 3 per order)</span></div>
            <asp:GridView ID="gvBulk" runat="server" AutoGenerateColumns="True"
                EmptyDataText="No bulk buyers found" Width="100%" />
        </div>

        <!-- Occasional Customers -->
        <div class="segment">
            <div class="segment-title"> Occasional Customers <span class="segment-
desc">(1 or 0 orders)</span></div>
            <asp:GridView ID="gvOccasional" runat="server"
AutoGenerateColumns="True"
                EmptyDataText="No occasional customers found" Width="100%" />
        </div>

        <!-- Low Value Customers -->
        <div class="segment">
            <div class="segment-title"> Low Value Customers <span class="segment-
desc">(Total spent less than $500)</span></div>
            <asp:GridView ID="gvLow" runat="server" AutoGenerateColumns="True"
                EmptyDataText="No low value customers found" Width="100%" />
        </div>

        <!-- Inactive Customers -->
        <div class="segment">
            <div class="segment-title"> Inactive Customers <span class="segment-
desc">(No orders in last 6 months)</span></div>
            <asp:GridView ID="gvInactive" runat="server" AutoGenerateColumns="True"
                EmptyDataText="No inactive customers found" Width="100%" />
        </div>

        <!-- Loyal Customers -->
        <div class="segment">
            <div class="segment-title"> Loyal Customers <span class="segment-
desc">(Frequent + Premium - More than 3 orders and spent > $2000)</span></div>
            <asp:GridView ID="gvLoyal" runat="server" AutoGenerateColumns="True"
                EmptyDataText="No loyal customers found" Width="100%" />
        </div>

        <div class="footer">
            <p>Data generated on: <asp:Label ID="lblGeneratedDate" runat="server"
/></p>

```

```

        <p>Based on order history from database</p>
    </div>

    <div class="button-group">
        <asp:Button ID="btnBack" runat="server" Text="← Back to Dashboard"
        CssClass="btn btn-back" OnClick="btnBack_Click" />
    </div>
</div>
</form>
</body>
</html>

```

CustomerSegmentation.aspx

```

using System;
using System.Data;
using System.Data.SqlClient;
using System.Configuration;

public partial class CustomerSegmentation : System.Web.UI.Page
{
    string connStr =
    ConfigurationManager.ConnectionStrings["PVFCCConnectionString"].ConnectionString;

    protected void Page_Load(object sender, EventArgs e)
    {
        if (!IsPostBack)
        {
            LoadAllSegments();
            lblGeneratedDate.Text = DateTime.Now.ToString("yyyy-MM-dd HH:mm:ss");
        }
    }

    protected void btnRefresh_Click(object sender, EventArgs e)
    {
        LoadAllSegments();
        lblGeneratedDate.Text = DateTime.Now.ToString("yyyy-MM-dd HH:mm:ss");
    }

    protected void btnBack_Click(object sender, EventArgs e)
    {
        Response.Redirect("Default.aspx");
    }

    private void LoadAllSegments()
    {
        try
        {
            using (SqlConnection con = new SqlConnection(connStr))
            {
                con.Open();

                // 1. PREMIUM CUSTOMERS who will Spent > $2000
                string sqlPremium = @"
                SELECT
                    C.Customer_Id AS [ID],
                    C.Customer_Name AS [Customer Name],
                    C.Customer_City AS [City],

```



```

        ISNULL(SUM(P.Standard_Price * OL.Ordered_Quantity), 0) AS [Total
Spent],
        COUNT(DISTINCT O.Order_Id) AS [Orders]
FROM CUSTOMER_t C
LEFT JOIN Users U ON C.Customer_Id = U.Customer_Id
LEFT JOIN Orders_t O ON U.UserID = O.UserID
LEFT JOIN Order_line_t OL ON O.Order_Id = OL.Order_Id
LEFT JOIN PRODUCT_t P ON OL.Product_Id = P.Product_Id
GROUP BY C.Customer_Id, C.Customer_Name, C.Customer_City
HAVING ISNULL(SUM(P.Standard_Price * OL.Ordered_Quantity), 0) > 2000
ORDER BY [Total Spent] DESC";
LoadData(con, sqlPremium, gvPremium);

```

```

// 2. FREQUENT CUSTOMERS places Orders > 3
string sqlFrequent = @"
SELECT
    C.Customer_Id AS [ID],
    C.Customer_Name AS [Customer Name],
    C.Customer_City AS [City],
    COUNT(DISTINCT O.Order_Id) AS [Orders],
    ISNULL(SUM(P.Standard_Price * OL.Ordered_Quantity), 0) AS [Total
Spent]

FROM CUSTOMER_t C
LEFT JOIN Users U ON C.Customer_Id = U.Customer_Id
LEFT JOIN Orders_t O ON U.UserID = O.UserID
LEFT JOIN Order_line_t OL ON O.Order_Id = OL.Order_Id
LEFT JOIN PRODUCT_t P ON OL.Product_Id = P.Product_Id
GROUP BY C.Customer_Id, C.Customer_Name, C.Customer_City
HAVING COUNT(DISTINCT O.Order_Id) > 3
ORDER BY [Orders] DESC";
LoadData(con, sqlFrequent, gvFrequent);

```

```

// 3. BULK BUYERS have Avg Order Quantity > 3
string sqlBulk = @"
SELECT
    C.Customer_Id AS [ID],
    C.Customer_Name AS [Customer Name],
    C.Customer_City AS [City],
    CAST(AVG(CAST(OL.Ordered_Quantity AS FLOAT)) AS DECIMAL(10,2))
AS [Avg Quantity],
    SUM(OL.Ordered_Quantity) AS [Total Items]
FROM CUSTOMER_t C
INNER JOIN Users U ON C.Customer_Id = U.Customer_Id
INNER JOIN Orders_t O ON U.UserID = O.UserID
INNER JOIN Order_line_t OL ON O.Order_Id = OL.Order_Id
GROUP BY C.Customer_Id, C.Customer_Name, C.Customer_City
HAVING AVG(CAST(OL.Ordered_Quantity AS FLOAT)) > 3
ORDER BY [Avg Quantity] DESC";
LoadData(con, sqlBulk, gvBulk);

```

```

// 4. OCCASIONAL CUSTOMERS palces order <= 1
string sqlOccasional = @"
SELECT
    C.Customer_Id AS [ID],
    C.Customer_Name AS [Customer Name],
    C.Customer_City AS [City],
    ISNULL(COUNT(DISTINCT O.Order_Id), 0) AS [Orders],
    ISNULL(SUM(P.Standard_Price * OL.Ordered_Quantity), 0) AS [Total
Spent]

FROM CUSTOMER_t C
LEFT JOIN Users U ON C.Customer_Id = U.Customer_Id

```

```

LEFT JOIN Orders_t O ON U.UserID = O.UserID
LEFT JOIN Order_line_t OL ON O.Order_Id = OL.Order_Id
LEFT JOIN PRODUCT_t P ON OL.Product_Id = P.Product_Id
GROUP BY C.Customer_Id, C.Customer_Name, C.Customer_City
HAVING ISNULL(COUNT(DISTINCT O.Order_Id), 0) <= 1
ORDER BY [Orders] DESC";
LoadData(con, sqlOccasional, gvOccasional);

// 5. LOW VALUE CUSTOMERS who will Spent < $500
string sqlLow = @"
SELECT
    C.Customer_Id AS [ID],
    C.Customer_Name AS [Customer Name],
    C.Customer_City AS [City],
    ISNULL(SUM(P.Standard_Price * OL.Ordered_Quantity), 0) AS [Total
Spent],
    ISNULL(COUNT(DISTINCT O.Order_Id), 0) AS [Orders]
FROM CUSTOMER_t C
LEFT JOIN Users U ON C.Customer_Id = U.Customer_Id
LEFT JOIN Orders_t O ON U.UserID = O.UserID
LEFT JOIN Order_line_t OL ON O.Order_Id = OL.Order_Id
LEFT JOIN PRODUCT_t P ON OL.Product_Id = P.Product_Id
GROUP BY C.Customer_Id, C.Customer_Name, C.Customer_City
HAVING ISNULL(SUM(P.Standard_Price * OL.Ordered_Quantity), 0) < 500
ORDER BY [Total Spent] ASC";
LoadData(con, sqlLow, gvLow);

// 6. INACTIVE CUSTOMERS places No orders in 6 months
string sqlInactive = @"
SELECT
    C.Customer_Id AS [ID],
    C.Customer_Name AS [Customer Name],
    C.Customer_City AS [City],
    CASE
        WHEN MAX(O.Order_Date) IS NULL THEN 'Never Ordered'
        ELSE CONVERT(VARCHAR, MAX(O.Order_Date), 106)
    END AS [Last Order],
    CASE
        WHEN MAX(O.Order_Date) IS NULL THEN 999
        ELSE DATEDIFF(MONTH, MAX(O.Order_Date), GETDATE())
    END AS [Months Inactive]
FROM CUSTOMER_t C
LEFT JOIN Users U ON C.Customer_Id = U.Customer_Id
LEFT JOIN Orders_t O ON U.UserID = O.UserID
GROUP BY C.Customer_Id, C.Customer_Name, C.Customer_City
HAVING MAX(O.Order_Date) IS NULL
    OR MAX(O.Order_Date) < DATEADD(MONTH, -6, GETDATE())
ORDER BY [Months Inactive] DESC";
LoadData(con, sqlInactive, gvInactive);

// 7. LOYAL CUSTOMERS are (Frequent + Premium)
string sqlLoyal = @"
SELECT
    C.Customer_Id AS [ID],
    C.Customer_Name AS [Customer Name],
    C.Customer_City AS [City],
    COUNT(DISTINCT O.Order_Id) AS [Orders],
    ISNULL(SUM(P.Standard_Price * OL.Ordered_Quantity), 0) AS [Total
Spent]
FROM CUSTOMER_t C
LEFT JOIN Users U ON C.Customer_Id = U.Customer_Id

```

```

        LEFT JOIN Orders_t O ON U.UserID = O.UserID
        LEFT JOIN Order_line_t OL ON O.Order_Id = OL.Order_Id
        LEFT JOIN PRODUCT_t P ON OL.Product_Id = P.Product_Id
        GROUP BY C.Customer_Id, C.Customer_Name, C.Customer_City
        HAVING COUNT(DISTINCT O.Order_Id) > 3
        AND ISNULL(SUM(P.Standard_Price * OL.Ordered_Quantity), 0) >
2000
        ORDER BY [Total Spent] DESC";
        LoadData(con, sqlLoyal, gvLoyal);

        UpdateStatistics(con);
    }
}
catch (Exception ex)
{
    Response.Write("<div style='color:red; padding:10px;'>Error: " + ex.Message
+ "</div>");
}
}

private void UpdateStatistics(SqlConnection con)
{
    try
    {
        SqlCommand cmdTotal = new SqlCommand("SELECT COUNT(*) FROM CUSTOMER_t",
con);
        int totalCustomers = Convert.ToInt32(cmdTotal.ExecuteScalar());
        lblTotalCustomers.Text = totalCustomers.ToString();

        SqlCommand cmdActive = new SqlCommand(@"
        SELECT COUNT(DISTINCT C.Customer_Id)
        FROM CUSTOMER_t C
        INNER JOIN Users U ON C.Customer_Id = U.Customer_Id
        INNER JOIN Orders_t O ON U.UserID = O.UserID", con);
        int activeCustomers = Convert.ToInt32(cmdActive.ExecuteScalar());
        lblActiveCustomers.Text = activeCustomers.ToString();
        lblInactiveCustomers.Text = (totalCustomers - activeCustomers).ToString();

        SqlCommand cmdRevenue = new SqlCommand(@"
        SELECT ISNULL(SUM(P.Standard_Price * OL.Ordered_Quantity), 0)
        FROM Order_line_t OL
        INNER JOIN PRODUCT_t P ON OL.Product_Id = P.Product_Id", con);
        decimal totalRevenue = Convert.ToDecimal(cmdRevenue.ExecuteScalar());
        lblTotalRevenue.Text = totalRevenue.ToString("N2");
    }
    catch (Exception ex)
    {
        System.Diagnostics.Debug.WriteLine("Stats error: " + ex.Message);
    }
}

private void LoadData(SqlConnection con, string query,
System.Web.UI.WebControls.GridView gv)
{
    try
    {
        SqlDataAdapter da = new SqlDataAdapter(query, con);
        DataTable dt = new DataTable();
        da.Fill(dt);
        gv.DataSource = dt;
        gv.DataBind();
    }
}

```

```

    }
    catch (Exception ex)
    {
        System.Diagnostics.Debug.WriteLine("Load error: " + ex.Message);
    }
}

```

TargetedMarketing.aspx + TargetedMarketing.aspx.cs(due to linking error same file)

```

<%@ Page Language="C#" AutoEventWireup="true" %>

<!DOCTYPE html>
<html>
<head runat="server">
    <title>Targeted Marketing</title>
    <style>
        body { font-family: Arial; margin: 20px; background: #f5f5f5; }
        .container { max-width: 1000px; margin: auto; background: white; padding: 20px;
border: 1px solid #ddd; }
        h2 { color: #333; border-bottom: 2px solid #4CAF50; padding-bottom: 10px; }
        .segment { margin-bottom: 20px; padding: 15px; border: 1px solid #ddd;
background: #fafafa; }
        .segment h3 { margin: 0 0 10px 0; color: #4CAF50; }
        .btn { padding: 8px 15px; margin: 5px; cursor: pointer; border: none;
background: #4CAF50; color: white; }
        .btn-back { background: #666; }
        .message { padding: 10px; margin: 10px 0; }
        .success { background: #e8f5e9; border: 1px solid #4CAF50; color: #2e7d32; }
        table { width: 100%; border-collapse: collapse; margin-top: 10px; }
        th { background: #f2f2f2; padding: 8px; text-align: left; border: 1px solid
#ddd; }
        td { padding: 8px; border: 1px solid #ddd; }
        .button-group { text-align: center; margin-top: 20px; }
    </style>
</head>
<body>
    <form id="form1" runat="server">
        <div class="container">
            <h2> Targeted Marketing</h2>

            <!-- Royal Customers -->
            <div class="segment">
                <h3> Royal Customers (Premium + Frequent)</h3>
                <p>Offer: Complimentary Gift + 25% Discount + Thank You Message</p>
                <asp:Button ID="btnRoyal" runat="server" Text="Send to Royal Customers"
CssClass="btn" OnClick="btnRoyal_Click" />
                <asp:Label ID="lblRoyal" runat="server" />
                <asp:GridView ID="gvRoyal" runat="server" AutoGenerateColumns="True"
Width="100%" EmptyDataText="No royal customers" />
            </div>

            <!-- Premium Customers -->
            <div class="segment">
                <h3> Premium Customers (Spent > $2000)</h3>

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        <p>Offer: 15% Discount + VIP Access</p>
        <asp:Button ID="btnPremium" runat="server" Text="Send to Premium
Customers" CssClass="btn" OnClick="btnPremium_Click" />
        <asp:Label ID="lblPremium" runat="server" />
        <asp:GridView ID="gvPremium" runat="server" AutoGenerateColumns="True"
Width="100%" EmptyDataText="No premium customers" />
    </div>

    <!-- Frequent Customers -->
    <div class="segment">
        <h3>Frequent Customers (Orders > 3)</h3>
        <p>Offer: Loyalty Points + Buy 3 Get 1 Free</p>
        <asp:Button ID="btnFrequent" runat="server" Text="Send to Frequent
Customers" CssClass="btn" OnClick="btnFrequent_Click" />
        <asp:Label ID="lblFrequent" runat="server" />
        <asp:GridView ID="gvFrequent" runat="server" AutoGenerateColumns="True"
Width="100%" EmptyDataText="No frequent customers" />
    </div>

    <!-- Inactive Customers -->
    <div class="segment">
        <h3>Inactive Customers (No orders in 6 months)</h3>
        <p>Offer: 20% Welcome Back Discount</p>
        <asp:Button ID="btnInactive" runat="server" Text="Send to Inactive
Customers" CssClass="btn" OnClick="btnInactive_Click" />
        <asp:Label ID="lblInactive" runat="server" />
        <asp:GridView ID="gvInactive" runat="server" AutoGenerateColumns="True"
Width="100%" EmptyDataText="No inactive customers" />
    </div>

    <!-- Low Value Customers -->
    <div class="segment">
        <h3>Low Value Customers (Spent < $500)</h3>
        <p>Offer: Free Shipping + First Purchase Discount</p>
        <asp:Button ID="btnLow" runat="server" Text="Send to Low Value
Customers" CssClass="btn" OnClick="btnLow_Click" />
        <asp:Label ID="lblLow" runat="server" />
        <asp:GridView ID="gvLow" runat="server" AutoGenerateColumns="True"
Width="100%" EmptyDataText="No low value customers" />
    </div>

    <!-- Bulk Buyers -->
    <div class="segment">
        <h3>Bulk Buyers (Avg Quantity > 3)</h3>
        <p>Offer: 30% Volume Discount + Wholesale Pricing</p>
        <asp:Button ID="btnBulk" runat="server" Text="Send to Bulk Buyers"
CssClass="btn" OnClick="btnBulk_Click" />
        <asp:Label ID="lblBulk" runat="server" />
        <asp:GridView ID="gvBulk" runat="server" AutoGenerateColumns="True"
Width="100%" EmptyDataText="No bulk buyers" />
    </div>

    <div class="button-group">
        <asp:Button ID="btnBack" runat="server" Text="← Back To DashBoard"
CssClass="btn btn-back" OnClick="btnBack_Click" />
    </div>
</div>
</form>

<script runat="server">

```

```
        string connStr =  
System.Configuration.ConfigurationManager.ConnectionStrings["PVFCCConnectionString"].Conn  
ectionString;
```

```
protected void Page_Load(object sender, EventArgs e)  
{  
    if (!IsPostBack)  
    {  
        LoadCustomers();  
    }  
}
```

```
private void LoadCustomers()  
{  
    LoadRoyal();  
    LoadPremium();  
    LoadFrequent();  
    LoadInactive();  
    LoadLowValue();  
    LoadBulk();  
}
```

```
private void LoadRoyal()  
{  
    string sql = @"  
        SELECT C.Customer_Name, C.Customer_City,  
                COUNT(O.Order_Id) AS Orders,  
                SUM(P.Standard_Price * OL.Ordered_Quantity) AS TotalSpent  
        FROM CUSTOMER_t C  
        JOIN Users U ON C.Customer_Id = U.Customer_Id  
        JOIN Orders_t O ON U.UserID = O.UserID  
        JOIN Order_line_t OL ON O.Order_Id = OL.Order_Id  
        JOIN PRODUCT_t P ON OL.Product_Id = P.Product_Id  
        GROUP BY C.Customer_Name, C.Customer_City  
        HAVING COUNT(O.Order_Id) > 3 AND SUM(P.Standard_Price *  
OL.Ordered_Quantity) > 2000";  
    LoadData(sql, gvRoyal);  
}
```

```
private void LoadPremium()  
{  
    string sql = @"  
        SELECT C.Customer_Name, C.Customer_City,  
                SUM(P.Standard_Price * OL.Ordered_Quantity) AS TotalSpent  
        FROM CUSTOMER_t C  
        JOIN Users U ON C.Customer_Id = U.Customer_Id  
        JOIN Orders_t O ON U.UserID = O.UserID  
        JOIN Order_line_t OL ON O.Order_Id = OL.Order_Id  
        JOIN PRODUCT_t P ON OL.Product_Id = P.Product_Id  
        GROUP BY C.Customer_Name, C.Customer_City  
        HAVING SUM(P.Standard_Price * OL.Ordered_Quantity) > 2000";  
    LoadData(sql, gvPremium);  
}
```

```
private void LoadFrequent()  
{  
    string sql = @"  
        SELECT C.Customer_Name, C.Customer_City, COUNT(O.Order_Id) AS Orders  
        FROM CUSTOMER_t C  
        JOIN Users U ON C.Customer_Id = U.Customer_Id  
        JOIN Orders_t O ON U.UserID = O.UserID
```

```

        GROUP BY C.Customer_Name, C.Customer_City
        HAVING COUNT(O.Order_Id) > 3";
        LoadData(sql, gvFrequent);
    }

    private void LoadInactive()
    {
        string sql = @"
            SELECT C.Customer_Name, C.Customer_City, MAX(O.Order_Date) AS LastOrder
            FROM CUSTOMER_t C
            JOIN Users U ON C.Customer_Id = U.Customer_Id
            LEFT JOIN Orders_t O ON U.UserID = O.UserID
            GROUP BY C.Customer_Name, C.Customer_City
            HAVING MAX(O.Order_Date) < DATEADD(MONTH, -6, GETDATE()) OR
MAX(O.Order_Date) IS NULL";
        LoadData(sql, gvInactive);
    }

    private void LoadLowValue()
    {
        string sql = @"
            SELECT C.Customer_Name, C.Customer_City,
                ISNULL(SUM(P.Standard_Price * OL.Ordered_Quantity), 0) AS
TotalSpent
            FROM CUSTOMER_t C
            JOIN Users U ON C.Customer_Id = U.Customer_Id
            LEFT JOIN Orders_t O ON U.UserID = O.UserID
            LEFT JOIN Order_line_t OL ON O.Order_Id = OL.Order_Id
            LEFT JOIN PRODUCT_t P ON OL.Product_Id = P.Product_Id
            GROUP BY C.Customer_Name, C.Customer_City
            HAVING ISNULL(SUM(P.Standard_Price * OL.Ordered_Quantity), 0) < 500";
        LoadData(sql, gvLow);
    }

    private void LoadBulk()
    {
        string sql = @"
            SELECT C.Customer_Name, C.Customer_City,
                AVG(CAST(OL.Ordered_Quantity AS FLOAT)) AS AvgQuantity
            FROM CUSTOMER_t C
            JOIN Users U ON C.Customer_Id = U.Customer_Id
            JOIN Orders_t O ON U.UserID = O.UserID
            JOIN Order_line_t OL ON O.Order_Id = OL.Order_Id
            GROUP BY C.Customer_Name, C.Customer_City
            HAVING AVG(CAST(OL.Ordered_Quantity AS FLOAT)) > 3";
        LoadData(sql, gvBulk);
    }

    protected void btnRoyal_Click(object sender, EventArgs e)
    {
        int count = gvRoyal.Rows.Count;
        if (count > 0)
            lblRoyal.Text = $" Offer sent to {count} Royal customers! Gift via
courier + 25% discount";
        else
            lblRoyal.Text = "No royal customers found.";
    }

    protected void btnPremium_Click(object sender, EventArgs e)
    {
        int count = gvPremium.Rows.Count;

```

```

        lblPremium.Text = count > 0 ? $" 15% discount sent to {count} Premium
customers!" : "No premium customers found.";
    }

    protected void btnFrequent_Click(object sender, EventArgs e)
    {
        int count = gvFrequent.Rows.Count;
        lblFrequent.Text = count > 0 ? $" Loyalty points offer sent to {count}
Frequent customers!" : "No frequent customers found.";
    }

    protected void btnInactive_Click(object sender, EventArgs e)
    {
        int count = gvInactive.Rows.Count;
        lblInactive.Text = count > 0 ? $" 20% welcome back discount sent to {count}
Inactive customers!" : "No inactive customers found.";
    }

    protected void btnLow_Click(object sender, EventArgs e)
    {
        int count = gvLow.Rows.Count;
        lblLow.Text = count > 0 ? $" Free shipping offer sent to {count} Low Value
customers!" : "No low value customers found.";
    }

    protected void btnBulk_Click(object sender, EventArgs e)
    {
        int count = gvBulk.Rows.Count;
        lblBulk.Text = count > 0 ? $" 30% volume discount sent to {count} Bulk
buyers!" : "No bulk buyers found.";
    }

    protected void btnBack_Click(object sender, EventArgs e)
    {
        Response.Redirect("Default.aspx");
    }

    private void LoadData(string sql, System.Web.UI.WebControls.GridView gv)
    {
        using (System.Data.SqlClient.SqlConnection con = new
System.Data.SqlClient.SqlConnection(connStr))
        {
            System.Data.SqlClient.SqlDataAdapter da = new
System.Data.SqlClient.SqlDataAdapter(sql, con);
            System.Data.DataTable dt = new System.Data.DataTable();
            da.Fill(dt);
            gv.DataSource = dt;
            gv.DataBind();
        }
    }
}
</script>
</body>
</html>

```